

Level 2 — Doing the Analysis

Hands-on clinical data analysis in R

Organisation: Neudata — #ClearDataClearImpact

> **Status:** Outline — materials to be developed after Level 1.

1. Why Level 2

Level 1 shows the whole research journey, and the analyses are run by the instructor while participants read and interpret the output. Level 2 is for the participants who want to **do the job themselves**. It goes deeper into a small number of methods that cover most clinical papers — describing data, linear regression and logistic regression — and asks, for each one: *What do the parameters mean? What are the assumptions? How do I check them? How do I interpret and report the results?* It gives the same care to the practical skills that make research possible: managing data properly, writing reproducible R scripts and producing publication-ready tables and figures.

By the end of Level 2, participants will be able to:

1. Organise an analysis as a reproducible R project (folders, scripts, Quarto report) that someone else could re-run.
2. Import, clean, recode and validate a clinical dataset, documenting every change in a cleaning log and a data dictionary.
3. Produce a publication-ready Table 1 and figures that meet journal requirements.
4. Fit, check, interpret and report linear and logistic regression models, including confounding, interaction, assumptions and diagnostics.
5. Handle missing data openly: describe it, run complete-case and multiple-imputation analyses, and report a sensitivity analysis.
6. Turn analysis output into a STROBE- or CONSORT-compliant Results section with tables and figures.

2. Who it is for and prerequisites

Participants	8–12 motivated staff from the client department (doctors, nurses, pharmacists) who want to analyse their own data.
Prerequisites	(1) Level 1 completed. (2) The free Neudata course Clinical Data Analysis in R , or equivalent basics: RStudio, objects and data frames, importing a CSV, basic tidyverse verbs, a simple plot. A short readiness check (30 minutes, self-marked) is sent before the course.
Equipment	Own laptop with R, RStudio and the course packages installed before Session 1 (installation guide and a 30-minute online set-up clinic).

Commitment	3 hours per session plus about 2–3 hours of homework per week; agreement from the line manager for protected time.
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3. Format

Sessions	8 × 3 hours over about 4–6 weeks (e.g. two sessions a week, or one per week with a one-week break mid-course).
Session rhythm	20–30 min concept recap → 2 h live coding and exercises (instructor codes, participants follow, then practise) → 20 min "my data" reflection and homework briefing. 15-minute break.
Shared dataset	FLUID-ICU (Course/Data/): the simulated ICU sepsis cohort (raw and clean data, data dictionary, cleaning log, SAP, R scripts) already used in Level 1. Participants rebuild the Level 1 results pack analyses themselves.
Own data	Each participant may bring a de-identified dataset (with the necessary ethics or data-protection approval). Two one-to-one slots of 30–45 minutes (after Sessions 3 and 6) help adapt the course scripts to their data.
Support	Shared course folder (scripts, solutions after each session), an online question channel, and an instructor "office hour" each week.
Main R packages	tidyverse (readr, dplyr, tidyr, ggplot2), here, janitor, readxl, gtsummary, flextable, broom, car, performance, pROC, mice, quarto/rmarkdown; optional: git via RStudio or GitHub Desktop.

4. Session-by-session outline

Session 1 — Reproducible projects: folders, scripts and reports

- **Concepts:** why reproducibility matters (the "re-run in a year" test); the analysis pipeline raw → clean → analysis → report.
- **Hands-on:** create an RStudio Project with a standard folder structure (`data/raw`, `data/clean`, `R/`, `output/`, `report/`); paths with `here`; script headers, comments and numbered scripts; packages loaded at the top; `sessionInfo()`.
- **Quarto/R Markdown:** a first report mixing text, code and output; rendering to Word and HTML; inline numbers in sentences.
- **FLUID-ICU:** set up the project and render a one-page "data overview" report.
- **Deliverable:** a working project folder and a rendered Quarto report.

Session 2 — Data management: import, cleaning, validation and documentation

- **Concepts:** raw data are never edited; the cleaning log; the data dictionary as the contract between data and analysis; tidy data.
- **Hands-on (FLUID-ICU raw file):** import CSV and Excel; check structure and types; find duplicates; recode sentinel values (999, -99) to missing; harmonise categories (M/Male/1);

parse dates; find impossible values with range and consistency checks; derive variables (cumulative fluid balance in ml/kg); set factor levels and reference categories.

- **Documentation:** write the cleaning log as you go; generate a data dictionary from the clean data; compare your clean file with the course clean file.
- **Version control basics:** dated versions and a change log at minimum; optional introduction to Git (commit, history, restore) through RStudio.
- **Deliverable:** `01_clean.R`, a cleaning log and a data dictionary for FLUID-ICU (and a first look at the participant's own data).

Session 3 — Describing data and a publication-ready Table 1

- **Concepts:** choosing the right summary (mean/SD vs median/IQR, n/%); distributions and skewness; missing values in tables; what Table 1 is for (and why p-values do not belong in Table 1 of an RCT).
- **Hands-on:** summaries with `dplyr`; Table 1 by exposure group with **gtsummary** (labels, units, statistics per variable type, missing counts, footnotes); export to Word with **flextable**; consistent decimals and journal style.
- **FLUID-ICU:** Table 1 by positive vs non-positive fluid balance, matching the Level 1 results pack.
- **Deliverable:** `02_describe.R` and Table 1 as a Word table. *One-to-one slot 1 follows this session.*

Session 4 — Publication-ready figures with ggplot2

- **Concepts:** principles of good scientific figures (show the data, one message per figure, honest axes, colour-blind-safe palettes, no 3-D or dynamite plots); which figure for which data.
- **Hands-on:** the grammar of graphics; histograms, box and dot plots, scatter plots with smoothers, bar charts of proportions with CIs, Kaplan–Meier curves (via a ready-made function), forest plots of model estimates; facets; a reusable course theme; labels with units; exporting at journal resolution (300–600 dpi; TIFF/PDF/PNG; exact width in mm) with `ggsave`.
- **FLUID-ICU:** a two-panel figure (fluid balance by outcome; mortality by fluid-balance group).
- **Deliverable:** two exported, journal-ready figures and the script that makes them.

Session 5 — Linear regression in depth

- **Concepts:** the model as a line; what the intercept and slope mean (in the units of the data); continuous and categorical predictors; centring and scaling; adjusted coefficients; confidence intervals and p-values for coefficients; R^2 .
- **Assumptions and diagnostics:** linearity, independence, constant variance, normality of residuals; influential points; residual and Q–Q plots (`performance::check_model, car`); multicollinearity (VIF).
- **When assumptions fail:** transformations (log of a skewed outcome and how to back-transform to ratios), non-linear terms (splines, introduced), robust alternatives and non-parametric comparisons.
- **Interaction in a linear model:** fitting, interpreting and plotting an interaction term.
- **Reporting:** a regression table with `broom/gtsummary`; the sentence "each 10 ml/kg higher fluid balance was associated with ...".
- **FLUID-ICU:** ventilator-free days and ICU length of stay among survivors (Questions 3 and 5).

- **Deliverable:** 04_linear.R, a diagnostics figure and a reporting table with a written interpretation.

Session 6 — Logistic regression in depth

- **Concepts:** odds and odds ratios; the logit link; what the coefficients mean; ORs for continuous predictors (per meaningful unit); crude vs adjusted ORs (confounding by indication made visible).
- **Model building:** choosing confounders from the question and a simple causal diagram (DAG), not from p-values; events per variable and overfitting; why stepwise selection misleads; confounding vs interaction (effect modification) and how to test and report an interaction.
- **Assumptions and checks:** linearity in the logit (splines or categorisation, compared); separation and sparse data; influential observations.
- **Performance basics:** discrimination (ROC curve, AUC with p_{ROC}) and calibration (calibration plot by risk groups); why these matter more for prediction than for explanation.
- **Reporting:** a table of crude and adjusted ORs with 95% CIs; a forest plot; STROBE wording.
- **FLUID-ICU:** positive fluid balance and 28-day mortality (Question 1), with the heart-failure interaction; AKI (Question 2).
- **Deliverable:** 05_logistic.R, a crude-vs-adjusted OR table, a forest plot and a short interpretation. *One-to-one slot 2 follows this session.*

Session 7 — Missing data in practice and sensitivity analyses

- **Concepts (recap from Level 1):** MCAR, MAR and MNAR; why complete-case analysis can mislead; what multiple imputation does and does not do.
- **Hands-on:** describe and visualise missingness (patterns, comparison of patients with and without missing lactate); complete-case analysis; multiple imputation with **mice** (choosing variables for the imputation model, including the outcome; number of imputations; checking convergence and imputed distributions; pooling with Rubin's rules); comparing complete-case and imputed estimates.
- **Sensitivity analyses:** what to vary (missing-data assumptions, outliers, definitions of exposure); how to report them briefly.
- **FLUID-ICU:** lactate and 28-day mortality (Question 4).
- **Deliverable:** 06_missing.R and a sensitivity-analysis table with two sentences of interpretation.

Session 8 — From output to manuscript

- **Concepts:** the reproducible report as the single source of every number; one home for each number; consistency between text, tables and figures.
- **Hands-on:** a Quarto report that runs the whole pipeline (clean → describe → model → missing data) and renders Word tables and figures in journal format; inline results in sentences; the participant flow diagram; the STROBE (or CONSORT) items for Methods and Results; writing the statistical-analysis paragraph and the Results text from the output.
- **Peer check:** participants swap reports and re-run each other's project from scratch ("does it reproduce?").
- **Final-project clinic:** plan and start the final project.
- **Deliverable:** a rendered report that reproduces all FLUID-ICU results from the raw file.

5. Per-session deliverables (summary)

Session	Deliverable
1	Project folder + rendered Quarto report
2	Cleaning script, cleaning log, data dictionary
3	Table 1 (Word) and script
4	Two journal-ready figures and script
5	Linear model script, diagnostics, reporting table and interpretation
6	Logistic model script, crude vs adjusted OR table, forest plot, interpretation
7	Missing-data script and sensitivity-analysis table
8	Full reproducible report from raw data to results

Each deliverable is submitted to the course folder before the next session; instructors give short written feedback.

6. Final project

Within **4 weeks after Session 8**, each participant (or pair) submits:

1. A **reproducible analysis**: a project folder that runs from the raw data to all tables and figures with one command (renders the Quarto report), including a cleaning log, a data dictionary and a short statistical analysis plan written before the main analysis.
2. A **draft Results section** (about 500–800 words) with Table 1, one main results table (regression) and one figure, written to the relevant reporting guideline, plus the statistical-analysis paragraph for the Methods.

The data can be **the participant's own** (preferred, with approval) or a **FLUID-ICU question** not used in their Level 1 group. A 10-minute presentation of the final project at a closing meeting is recommended.

7. Assessment

Component	Weight	How it is assessed
Session deliverables (8)	40%	Completed and running (pass/fail per deliverable, with short feedback)
Final project — reproducibility	20%	An instructor re-runs the project on a clean computer; all results reproduce
Final project — analysis	20%	Appropriate methods, assumptions checked, missing data handled, interpretation correct
Final project — reporting	20%	Publication-ready tables and figures; Results text with effect sizes and 95% CIs; guideline items covered; calibrated language

Marks are used for feedback; there is no ranking.

8. Materials to be developed

Material	Content
Slides (EN + VN)	8 short concept decks (20–30 slides each) with speaker notes
R scripts	Session starter scripts (with gaps) and complete instructor scripts, built from the Level 1 FLUID-ICU scripts in <code>Course/Data/R/</code>
Exercises	One exercise sheet per session (core + stretch tasks) on FLUID-ICU, with a "your data" task
Solutions	Full worked solutions and rendered outputs, released after each session
Quarto templates	Report template (Word output in journal style), Table 1 and regression-table helpers, a ggplot2 course theme
Checklists	Reproducibility checklist; regression assumptions checklist; Results-section checklist (STROBE/CONSORT)
Set-up guide	Installation guide, package list and readiness check
Assessment	Deliverable checklists and the final-project rubric